

ON LACUNARY TRIGONOMETRIC SERIES

BY R. SALEM AND A. ZYGMUND

THE MASSACHUSETTS INSTITUTE OF TECHNOLOGY AND THE UNIVERSITY OF CHICAGO

Communicated July 21, 1947

Let us consider a lacunary trigonometric series

$$\sum_{k=1}^{\infty} (a_k \cos n_k x + b_k \sin n_k x), \text{ with } n_{k+1}/n_k > q > 1, \quad (1)$$

and let us confine our attention to the interval $0 \leq x \leq 2\pi$. Let $S_N(x)$ denote the N th partial sum of (1), that is to say the sum of the terms with $k = 1, 2, \dots, N$. Let $C_N = \{1/2(a_1^2 + b_1^2 + \dots + a_N^2 + b_N^2)\}^{1/2}$. In a recent note,¹ J. Ferrand and R. Fortet considered the behavior of the ratio $S_N(x)/C_N$. They stated, without proof, that if $C_N \rightarrow +\infty$ as $N \rightarrow \infty$, the distribution function of $S_N(x)/C_N$ tends to the Gaussian distribution with mean value 0 and dispersion 1. Clearly, the result as stated cannot be correct since, if $c_k = (a_k^2 + b_k^2)^{1/2}$ increases very rapidly, the distribution function tends to that of cosine. Thus some kind of restriction on the c_k is necessary. (Mr. Erdős informs us that a proof of the result, under the condition $C_n \rightarrow \infty$, $c_n = O(1)$, will be published in a paper written jointly by him, Mlle. Ferrand, Fortet and Kac.) In this note we propose to give a complete solution of the problem.

By a distribution function we shall mean any non-decreasing and continuous to the right function $F(y)$ with $F(-\infty) = 0$, $F(+\infty) = 1$. Let $Z_n(y)$ be the set of points x from $(0, 2\pi)$ at which $S_N(x)/C_N \leq y$. Let $F_N(y) = |Z_N(y)|/2\pi$ (by $|E|$ we mean the measure of the set E), so that F_N is the distribution function of S_N/C_N .

(i) If $F_N(y)$ tends to a distribution function $F(y)$ (at the points of continuity of the latter) such that either $F(y) > 0$ or $F(y) < 1$ for all finite y , then

$$c_n/C_n \rightarrow 0. \quad (2)$$

(ii) If (2) is satisfied and if $C_n \rightarrow \infty$, then $F_N(y)$ tends to the Gaussian distribution with mean value 0 and dispersion 1.

We shall even prove a more general result, namely, that the distribution function of $S_N(x)/C_N$ on every fixed set of positive measure tends to the Gaussian distribution.

(iii) Let E be a point set on $(0, 2\pi)$, with $|E| > 0$, and let $F_N(y; E) = |Z_N(y) \cap E|/|E|$. If $C_n \rightarrow \infty$ and if (2) holds, then $F_N(y; E)$ tends to the Gaussian distribution with mean value 0 and dispersion 1.

To prove (i), let us assume for example that $F(y) < 1$ for all finite y , and that (2) is false. Hence $c_n/C_n > \sqrt{2}\epsilon > 0$ for infinitely many N .

Let us confine our attention to such N . Obviously, $C_{N-1}/C_N < (1 - \epsilon^2)^{1/2}$, and the formula,

$$\frac{S_N}{C_N} = \frac{S_{N-1}}{C_{N-1}} \cdot \frac{C_{N-1}}{C_N} + \frac{a_N \cos n_N x + b_N \sin n_N x}{C_N},$$

shows that at every point x from $Z_{N-1}(y)$, $y > 0$, we have $S_N(x)/C_N \leq y(1 - \epsilon^2)^{1/2} + \sqrt{2}$. It follows that $Z_{N-1}(y)$ is included in $Z_N(y(1 - \epsilon^2)^{1/2} + \sqrt{2})$, and so $F_{N-1}(y) \leq F_N(y(1 - \epsilon^2)^{1/2} + \sqrt{2})$. Let y be a point of continuity of F . Making $N \rightarrow \infty$, we get $F(y) \leq F(y(1 - \epsilon^2)^{1/2} + \sqrt{2})$. However, from the assumption that $F(y)$ is always < 1 we obtain that there are points $y > 0$ of continuity of F , at which $F(y(1 - \epsilon^2)^{1/2} + \sqrt{2}) < F(y)$. This contradiction completes the proof of (i). Condition $C_N \rightarrow \infty$ and condition (2) imply that $\text{Max}_{1 \leq k \leq N} c_k / C_N \rightarrow 0$ as $N \rightarrow \infty$. This remark will be used repeatedly below.

To prove (iii), let us assume for the sake of brevity that (1) is a purely cosine series, and let us write A_N for C_N (in the general case we merely represent the terms of (1) in the form $c_k \cos(n_k x + \theta_k)$ and the argument that follows remains unchanged). It is enough to prove that over any finite range of λ the characteristic function of $F_N(y; E)$ uniformly approaches that of the Gaussian distribution. The characteristic function of $F_N(y; E)$ is

$$\begin{aligned} \int_{-\infty}^{+\infty} e^{i\lambda y} dF_N(y; E) &= |E|^{-1} \int_E e^{i\lambda S_N(x)/A_N} dx = \\ &= |E|^{-1} \int_E \exp. \left\{ i\lambda A_N^{-1} \sum_{k=1}^N a_k \cos n_k x \right\} dx = \\ &= |E|^{-1} \int_E e^{o(1)} \prod_{k=1}^N \left(1 + \frac{i\lambda a_k}{A_N} \cos n_k x \right) \exp. \left\{ -\frac{1}{2} \frac{\lambda^2 a_k^2}{A_N^2} \cos^2 n_k x \right\} dx, \quad (3) \end{aligned}$$

on account of (2) and of the relation $\exp. z = (1 + z) \exp. \{ \frac{1}{2} z^2 + o(|z|^2) \}$ valid for $z \rightarrow 0$. Due to (2) and $A_N \rightarrow \infty$, the term $o(1)$ in $e^{o(1)}$ tends to 0 uniformly in x as $N \rightarrow \infty$ (provided $\lambda = O(1)$, an assumption we always make from now on). Let us now observe that

$$\left| \prod_{k=1}^N \left(1 + \frac{i\lambda a_k}{A_N} \cos n_k x \right) \right| \leq \prod_{k=1}^N \left(1 + \frac{\lambda^2 a_k^2}{A_N^2} \right)^{1/2} \leq e^{\gamma^2}, \quad (4)$$

and that

$$\sum_{k=1}^N \frac{a_k^2}{A_N^2} \cos^2 n_k x = 1 + \sum_{k=1}^N \frac{a_k^2}{2A_N^2} \cos 2n_k x = 1 + \xi_N(x).$$

The measure of the set of points where $|\xi_N(x)| \geq \delta > 0$ is less than $\delta^{-2} \int_0^{2\pi} \xi_N^2(x) dx = \frac{1}{4\pi} \delta^{-2} (a_1^4 + \dots + a_N^4) / A_N^4$, and so tends to 0 on account of (2) and $A_N \rightarrow \infty$. Since $|\xi_N(x)| \leq 1$, it follows that, with an error tending uniformly to 0 as $N \rightarrow \infty$, the integral (3) is

$$|E|^{-1} e^{-\lambda^2/2} \int_E \prod_{k=1}^N \left(1 + \frac{i\lambda a_k}{A_N} \cos n_k x\right) dx = |E|^{-1} e^{-\lambda^2/2} I_N, \quad (5)$$

and it is enough to prove that I_N tends uniformly to $|E|$. We shall write

$$\prod_1^N \left(1 + i\lambda \frac{a_k}{A_N} \cos n_k x\right) = \alpha_0^{(N)} + \sum_1 \alpha_\gamma^{(N)} \cos \gamma x, \quad \epsilon_k = \epsilon_k^{(N)} = |i\lambda a_k/A_N|, \quad (6)$$

so that $\alpha_\gamma^{(N)}$, ϵ_k depend on λ also.

If $q \geq 3$, the proof of $I_N \rightarrow |E|$ is simple. For the numbers $\alpha_\gamma^{(N)}$ which actually occur in (6) correspond to the indices $\gamma \geq 0$ of the form $\pm n_{k_1} \pm n_{k_2} \pm \dots \pm n_{k_p}$ ($k_1 > k_2 > \dots > k_p$). For $q \geq 3$ such a representation (if it exists) is unique. In particular, $\alpha_0^{(N)} = 1$. Thus $I_N = |E| + \pi \sum h_\gamma \alpha_\gamma^{(N)}$, where h_γ are the cosine Fourier coefficients of the characteristic function of E . On account of (2), each $\alpha_\gamma^{(N)}$ tends to 0 as $N \rightarrow \infty$ and $\gamma = 1, 2, \dots$. Thus, if γ_0 is fixed,

$$\sum_1 |h_\gamma \alpha_\gamma^{(N)}| \leq \sum_1^{\gamma_0} + \sum_{\gamma_0+1}^{\infty} \leq o(1) + \left(\sum_{\gamma_0+1}^{\infty} h_\gamma^2\right)^{1/2} \left(\sum_{\gamma_0+1}^{\infty} |\alpha_\gamma^{(N)}|^2\right)^{1/2}. \quad (7)$$

The first factor in the last product is arbitrarily small if γ_0 is large enough (since $\sum h_\gamma^2 < \infty$), and the last factor is bounded by, (6), (4), and Parseval's formula. Thus $I_N \rightarrow |E|$, and (iii) follows. (If $E = (0, 2\pi)$, then even the weaker condition $q \geq 2$ implies that $\alpha_0^{(N)} = 1$, so that $I_N = 2\pi$, and (ii) follows.)

The proof of the general case of (iii) is based on two lemmas.

LEMMA 1. *The number of solutions of $n_{k_1} + n_{k_2} + \dots + n_{k_p} = a$, where a is given is less than C^{p-1} , where $C = C(q)$ depends on q only.* For assuming that $k_1 > k_2 > \dots > k_p$, we have $n_{k_1} \leq a \leq n_{k_1} q/(q-1)$, so that n_{k_1} takes $\leq C(q)$ values. The same applies to n_{k_2} when n_{k_1} is fixed and so on.

LEMMA 2. *The number of solutions of $n_{k_1} + \dots + n_{k_p} - n_{k_{p+1}} = a$ ($k_1 > k_2 > \dots > k_p > k_{p+1}$) is less than C^p , where C is the same as in Lemma 1.* For $p = 1$ this is immediate since $n_{k_1} - n_{k_2} = a$ implies $n_{k_1}(1 - 1/q) \leq a \leq n_{k_1}$, and the ratio of the extreme terms here is the same as in the inequality for a in the proof of Lemma 1. Combining this special case with Lemma 1 we get Lemma 2.

Let us now revert to the first formula (6). As the argument (7) shows, (iii) will be established when we show that $\alpha_0^{(N)} \rightarrow 1$, $\alpha_\gamma^{(N)} \rightarrow 0$ for $\gamma = 1, 2, \dots$. As a matter of fact, we can prove that $\alpha_\gamma^{(N)} \rightarrow 0$ uniformly in $\gamma > 0$. The γ actually occurring in (6) are ≥ 0 and of the form $\pm n_{k_1} \pm n_{k_2} \pm \dots \pm n_{k_p}$ ($k_1 > k_2 > \dots > k_p$). Let us fix p and let us write the last equation in the form

$$n_{h_1} + n_{h_2} + \dots + n_{h_s} = \gamma + n_{i_1} + n_{i_2} + \dots + n_{i_t}. \quad (8)$$

with $p = s + t$, $h_1 > h_2 > \dots > h_s$, $l_1 > l_2 > \dots > l_t$. Let us also fix s and t . Clearly, the case $s = t = 0$ leads to the contribution 1 in $\alpha_0^{(N)}$ so that, in order to estimate $\alpha_0^{(N)} - 1$, $\alpha_\gamma^{(N)}$ ($\gamma = 1, 2, \dots$) we must assume that $s \geq 1$, $t \geq 0$. For the clarity of presentation, let us first assume that $s > 1$, $t > 1$. Let us also assume that $h_s < l_t$. Thus we investigate the solutions of (8) under the conditions

$$\gamma \geq 0, s > 1, t > 1 - \text{fixed}, h_s < l_t. \quad (9)$$

Every solution of (8) contributes to $\alpha_\gamma^{(N)}$ (or to $\alpha_0^{(N)} - 1$) as much as $\epsilon_{h_1} \dots \epsilon_{h_s} \epsilon_{l_1} \dots \epsilon_{l_t} / 2^{p-1}$ in absolute value. The total contribution of the solutions of (8) satisfying (9) is therefore $\leq S = 2^{-(p-1)} \sum \epsilon_{h_1} \dots \epsilon_{h_s} \epsilon_{l_1} \dots \epsilon_{l_t}$, summation being extended over all indices satisfying (8) and (9). Let $\eta = \max. (\epsilon_1, \epsilon_2, \dots, \epsilon_N)$ so that $\eta = \eta_{N,\lambda} \rightarrow 0$ as $N \rightarrow \infty$, $\lambda = O(1)$. Then $S \leq 2^{-(p-1)} \eta \sum \epsilon_{h_1} \dots \epsilon_{h_{s-1}} \epsilon_{l_1} \dots \epsilon_{l_t}$ where summation is extended over the same indices $h_1, \dots, h_s, l_1, \dots, l_t$ as before. It follows that

$$S \leq 2^{-(p-1)} \eta (\sum \epsilon_{h_1}^2 \dots \epsilon_{h_{s-1}}^2)^{1/2} (\sum \epsilon_{l_1}^2 \dots \epsilon_{l_t}^2)^{1/2} \quad (10)$$

In $\sum \epsilon_{h_1}^2 \dots \epsilon_{l_t}^2$ each term must be taken as many times as for given $l_1, l_2, \dots, l_t$ one can find various groups of $h_1, h_2, \dots, h_s$ satisfying (8). By (8) and Lemma 1, this sum is $\leq C^{s-1} \sum^* \epsilon_{h_1}^2 \dots \epsilon_{h_{s-1}}^2$, the star meaning that now each term occurs once only. Since $\sum^* \epsilon_{h_1}^2 \dots \epsilon_{h_{s-1}}^2 \leq (\epsilon_1^2 + \dots + \epsilon_N^2)^{s-1} / (s-1)! = (2\lambda^2)^{s-1} / (s-1)!$, we get $\sum \epsilon_{h_1}^2 \dots \epsilon_{h_{s-1}}^2 \leq 2^s C^{s-1} \lambda^{2(s-1)} / (s-1)!$. Similarly, in the sum $\sum \epsilon_{h_1}^2 \dots \epsilon_{h_{s-1}}^2$ in (10) each term must be taken as many times as for given $h_1, h_2, \dots, h_{s-1}$ one can find various groups of $l_1, \dots, l_t$ satisfying (8). Since $h_s < l_t$, an application of Lemma 2 gives

$$\begin{aligned} \sum \epsilon_{h_1}^2 \dots \epsilon_{h_{s-1}}^2 &\leq C^t \sum^* \epsilon_{h_1}^2 \dots \epsilon_{h_{s-1}}^2 = (\epsilon_1^2 + \dots + \epsilon_N^2)^{s-1} C^t / (s-1)! \\ &= C^t (2\lambda^2)^{s-1} / (s-1)! \end{aligned}$$

Hence

$$S \leq \eta (\lambda^2 C / 2)^{(p-1)/2} / \{(s-1)!\}^{1/2} \leq \eta (\lambda^2 C / 2)^{(p-1)/2} / \{[1/2(p-1)]!\}^{1/2}.$$

If we drop the last condition in (9), we have to double this estimate. If we vary s, t but keep $p = s + t$ fixed, the result will be less than p times the last expression. Finally, summing over p and observing that the series with terms $(\lambda^2 C / 2)^{(p-1)/2} p / \{[1/2(p-1)]!\}^{1/2}$ is uniformly convergent in $\lambda = O(1)$, we see that the total contribution of the terms so far considered is $\leq H\eta$, where H is uniformly bounded for $\lambda = O(1)$. If we obtain a similar estimate for the terms so far disregarded, (iii) will be established.

But $s = 0$ implies $t = 0$ and can be omitted. If $t = 0$, we argue as before (by majorizing ϵ_{h_s} by η). The case $s = 1$ with $h_1 < l_t$ leads to $\gamma + n_{h_1} + \dots + n_{l_t} - n_{h_1} = 0$ and is impossible; if $l_t < h_1$, the treatment is

as above. Finally, if $t = 1$ ($s > 1$), and $n_{h_1} + \dots + n_{h_s} = \gamma + n_{h_s}$, the case $h_s < l_t$ offers nothing new; if $h_s > l_t$, then $S \leq \eta 2^{-(p-1)} \sum \epsilon_{h_1} \dots \epsilon_{h_s} \leq \eta 2^{-(p-1)} \{C^s \sum \epsilon_{h_1}^2 \dots \epsilon_{h_s}^2\}^{1/2}$, and the argument concludes as before. Thus (iii) is proved.

Suppose now that (1) is of the class L^2 . The remainder $(a_N \cos n_N x + b_N \sin n_N x) + \dots$ of (1) represents then a certain function $R_N(x)$. By $G_N(y)$ we shall mean the distribution function of $R_N(x)/D_N$, where $D_N = \{1/2(a_N^2 + b_N^2) + \dots\}^{1/2}$. The proofs of the following two results are repetitions of those of (i) and (iii).

(iv) If (1) is of the class L^2 , and if $G_N(y)$ tends to the distribution function $G(y)$ such that either $G(y) > 0$ or $G(y) < 1$ for all finite y , then $c_N/D_N \rightarrow 0$.

(v) Suppose that (1) is of the class L^2 , that $c_N/D_N \rightarrow 0$, and let E be a fixed set of positive measure. The distribution function of $R_N(x)/D_N$ relative to E tends to the Gaussian distribution with mean value 0 and dispersion 1.

Theorems (iii) and (v) have analogs for power series (2)

$$\sum_{k=1}^{\infty} c_k e^{i n_k x} \quad (n_{k+1}/n_k > q > 1) \quad (11)$$

We shall write $C_N = \{1/2(|c_1|^2 + \dots + |c_N|^2)\}^{1/2}$, $D_N = \{1/2(|c_N|^2 + \dots)\}^{1/2}$. The real and the imaginary part of the N th partial sum of (11) will be denoted by $U_N(x)$, $V_N(x)$, and of the N th remainder of (11) (if $C_N = O(1)$) by $U'_N(x)$, $V'_N(x)$, respectively.

(vi) Let E be a fixed set in $(0, 2\pi)$ of positive measure. If $C_N \rightarrow \infty$, $c_N = o(C_N)$, then the two dimensional distribution function, relative to E , of the pair of functions $U_N(x)/C_N$, $V_N(x)/C_N$ tends to $(2\pi)^{-1} \int_{-\infty}^{\xi} \int_{-\infty}^{\eta} e^{-1/2(\lambda^2 + \mu^2)} d\lambda d\mu$. The same conclusion holds for $U'_N(x)/D_N$, $V'_N(x)/D_N$ if (1) is of the class L^2 and if $c_N = o(D_N)$.

It is enough to sketch the proof of the first part of (vi). Let $Z_N(\xi, \eta)$ denote the set of points x such that $U_N(x)/C_N \leq \xi$, $V_N(x)/C_N \leq \eta$, and let $F_N(\xi, \eta) = |Z_N(\xi, \eta)E|/|E|$. Let $c_k = |c_k|e^{i\alpha_k}$. The characteristic function of F_N is

$$\begin{aligned} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{i(\lambda\xi + \mu\eta)} dF_N(\xi, \eta) &= |E|^{-1} \int_E e^{i(\lambda U_N + \mu V_N)/C_N} dx = \\ &= |E|^{-1} \int_E \exp. i C_N^{-1} \left\{ \sum |c_k| (\lambda \cos(n_k x + \alpha_k) + \mu \sin(n_k x + \alpha_k)) \right\} dx = \\ &= |E|^{-1} \int_E \exp. \{i C_N^{-1} (\lambda^2 + \mu^2)^{1/2} \sum |c_k| \cos(n_k x + \alpha'_k)\} dx \end{aligned}$$

where the α'_k depend now on λ, μ . As $N \rightarrow \infty$, and with an error $o(1)$, the last expression is

$$\begin{aligned} e^{-1/2(\lambda^2 + \mu^2)} |E|^{-1} \int_E \prod_{k=1}^N \{1 + i C_N^{-1} (\lambda^2 + \mu^2)^{1/2} |c_k| \cos(n_k x + \alpha'_k)\} dx \\ \rightarrow e^{-1/2(\lambda^2 + \mu^2)} \end{aligned}$$

which completes the proof.

The main idea used throughout this paper can be applied to other cases,

for example to linear combinations of the partial sums (or the remainders) of (1) and (11). We can also study the distribution functions in $(-\infty, +\infty)$ of the partial sums and of the remainders of the series $\sum c_k e^{i p_k t}$ with linearly independent exponents. We obtain the results parallel to the ones given above but with easier proofs.³ The infinite products $\prod(1 + i\alpha_k \cos n_k x)$ can also be used to obtain a simple proof of the theorem of Banach asserting the existence of continuous functions with prescribed Fourier coefficients at lacunary places. To these and other problems we shall return elsewhere.

¹ *Comptes Rendus Paris Acad. Sci.*, 224, 516-578 (1947).

² See also *loc. cit.* (1).

³ See also Kac and Steinhaus, *Studia Math.*, 7, 1-15 (1938).

TRANSFORMATION THEORY OF PHYSICAL CURVES

BY EDWARD KASNER AND JOHN DE CICCIO

DEPARTMENTS OF MATHEMATICS, COLUMBIA UNIVERSITY, NEW YORK, AND ILLINOIS INSTITUTE OF TECHNOLOGY, CHICAGO

Communicated August 5, 1947

1. The importance of projective transformations¹ in dynamics was brought out by Appell in 1889. If (ϕ, ψ) are the rectangular components of any positional field of force in the plane, the corresponding equations of motion of a particle of unit mass are

$$\frac{d^2x}{dt^2} = \phi(x, y), \quad \frac{d^2y}{dt^2} = \psi(x, y). \quad (1)$$

If an arbitrary point transformation, unaccompanied by any change in the time, is applied, the new differential equations will usually involve not only x and y , but also the velocity components dx/dt , dy/dt . In fact, the only exception is where the point transformation is merely affine.

Appell showed that if a general collineation

$$X = \frac{a_1x + b_1y + c_1}{a_0x + b_0y + c_0}, \quad Y = \frac{a_2x + b_2y + c_2}{a_0x + b_0y + c_0}, \quad (2)$$

is accompanied by a change of the time of the form

$$dT = \frac{dt}{k(a_0x + b_0y + c_0)^2} \quad (3)$$

the new differential equations will be of the original form